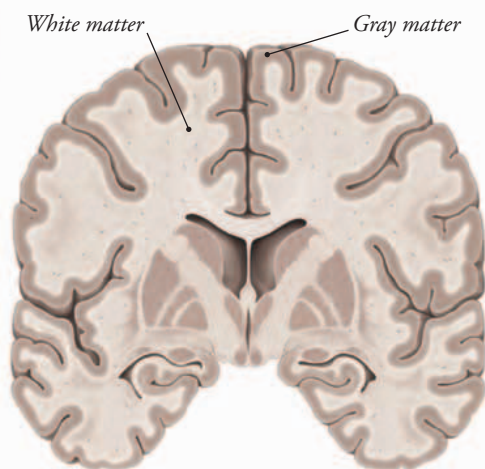


Gray matter

The largest part of your brain is called the cerebrum. Its wrinkly outer surface is made up of a thin layer of nerve cells known as gray matter, or the cerebral cortex. This layer is vital, because it is where you feel, see, hear, think, imagine, and remember. It also controls all your body movements.

WHITE MATTER

Beneath the thin outer layer of gray matter lies the paler white matter. This is packed with nerve fibers that carry signals between different parts of the brain, and between the brain and the rest of the body.



FAST FACTS

- There are more than 125 trillion (125,000,000,000,000) connections between your brain cells.
- Your brain is only 2 percent of your weight, but it uses around 20 percent of your body's energy.
- The cerebrum takes up two-thirds of the space in your skull.

If your wrinkly cerebral cortex was spread out flat, it would cover the same area as a pillowcase.

This area is associated with thinking, imagination, and personality.

Known as Broca's area, this region controls speech.

This area controls complex movements.

Signals received from the ears are turned into sounds here.

Sounds, and where they come from, are identified here.

MAPPING THE MIND

Different parts of the cerebral cortex specialize in different jobs. Scientists discovered this by studying people with brain damage in particular parts of the cortex. People with damage in a part of the brain called Broca's area, for example, lose the ability to speak, so Broca's area must be important in controlling language.